MILITARY SPECIFICATION

HEATER, ELECTRIC, IMMERSION
(Applicable to A/M32R-6 Recharger)

1. SCOPE

1.1 Scope. This specification covers an immersion heater rated at 9.5 KW, 220/440 V.A.C.. The heater is intended for use with the A/M32R-6 Oxygen/ Nitrogen Recharger.

1.2 Classification. The immersion heater shall be of one type and size.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-636	Boxes, Fiber
PPP-B-640	Boxes, Folding, Fiberboard, Corrugated, Triple-Wall

Military

MIL-P-116	Preservation, Methods of
MIL-L-10547	Liners, Case, and Sheet, Overwrap, Water-Vaporproof, or Waterproof, Flexible

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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NATIONAL BUREAU OF STANDARDS (NBS)

Handbook H28	Screw Thread Standards for Federal Services
Handbook H30	National Electric Safety Code

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

UNDERWRITERS' LABORATORIES, INC.

UL499	Standards for Electrical Heating Appliances

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., Avenue of the Americas, New York, N.Y. 10013; 207 East Ohio St., Chicago, Ill. 60611; or 500 Sansome St., San Francisco, Ca. 94111.)

3. REQUIREMENTS

3.1 First article. This specification makes provision for first article testing. The supplier shall furnish two complete immersion heaters for first article tests.

3.2 Design. The immersion heater shall be of one size and type and shall conform to size and tolerances as specified in Figure 1. The heater shall be rated at 9.5 KW.

3.2.1 Power supply. The heater shall be designed to operate on 220/440 volt, 60 hertz, 1 phase power supply.

3.2.2 Safety. The immersion heater shall be designed to provide maximum safety to personnel and equipment.

3.2.3 Material. All material used in the manufacture of the heaters shall be new and of the quality necessary to produce a heater to meet the requirements herein.

3.2.4 Components. The heater shall consist of the following components.

3.2.4.1 Envelope. The envelope, or tube, shall be of aluminum, either solid cast about the heating elements or, it may be a tube solidly packed with a high thermal conductive, non-hydroscopic material to provide maximum heat transfer to the outer surface of the tube. The tube shall conform to dimensions shown in Figure 1.

3.2.4.2 Mounting flange. The flange shall be of aluminum plate and shall conform with a configuration shown in Figure 1. The mounting flange shall be designed in such a manner as to prevent leakage between the flange, heater tube and envelope or tube.

3.2.5 Electrical. All electrical components, parts, insulation and circuits comprising the unit shall meet the requirements of National Bureau of Standards Handbook H30 and shall be suitable for the use intended in respect to heat resistance and moisture absorption.

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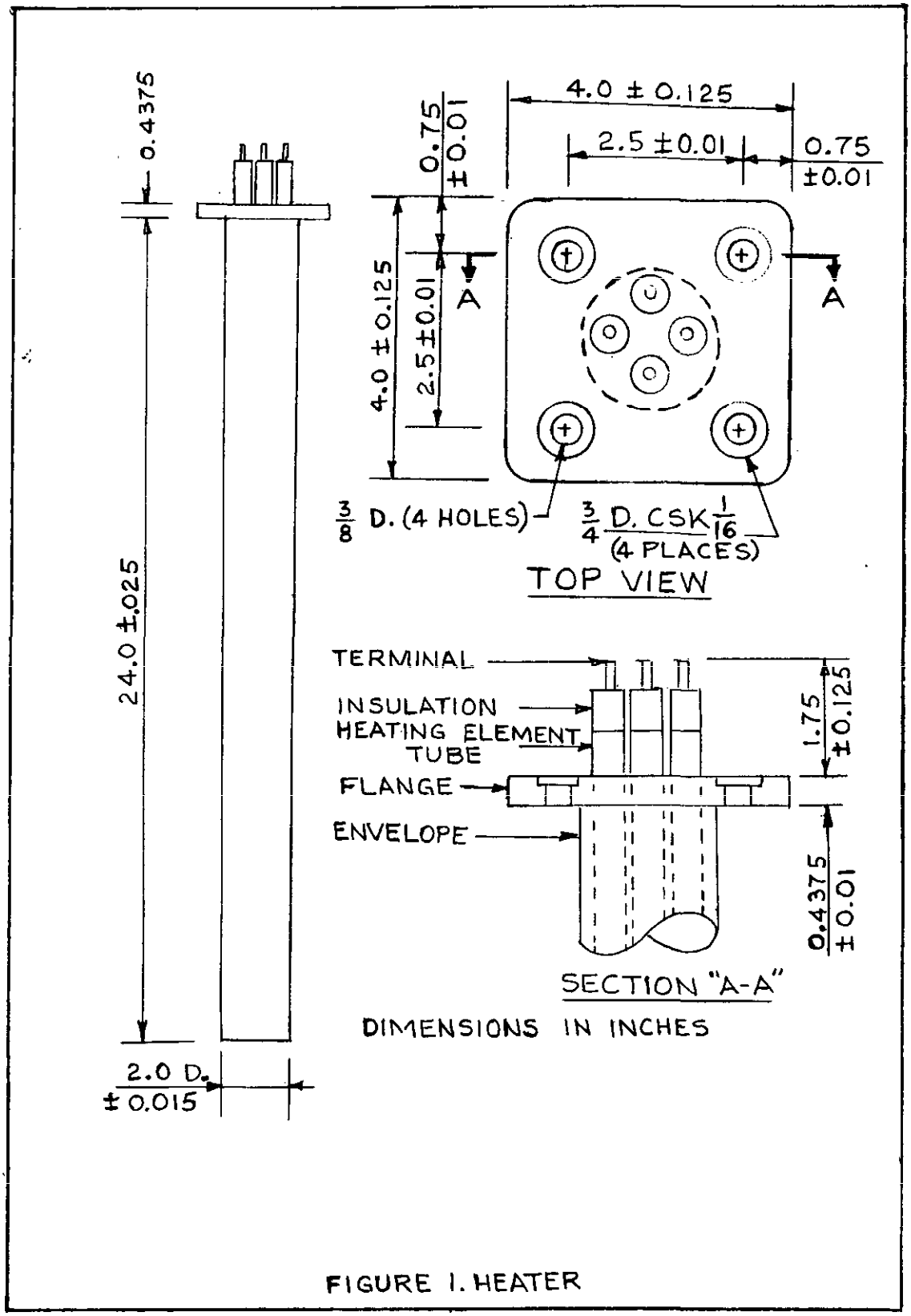


FIGURE 1. HEATER

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3.2.5.1 Heating elements. Heating elements shall be fabricated from high resistance alloy suitable for use in immersion heaters and shall conform to Underwriters' Laboratory Standard UL499. The heating element, or wire coil, shall be encased in a metallic tube and electrically insulated from the tube by high dielectric, high thermal, conductive material with low moisture absorbent qualities. The tube shall be enclosed in the envelope specified in para 3.2.4.1.

3.2.5.2 Electrical terminals. The electrical terminals shall be designed to give maximum possible clearance between live parts and between live parts and ground. The design shall prevent absorption or leakage of moisture from the environment to the inside of the heater. Terminal sizes shall be sufficiently large to carry the maximum current of the heater without overheating. Screw threads shall be in accordance with National Bureau of Standards Handbook H28. The terminals shall be rigidly fastened to prevent them from twisting when the terminal nut is tightened.

3.3 Construction. The immersion heaters shall be constructed so that when installed in an A/M32R-6 Oxygen/Nitrogen recharger and connected to the power supply, they will perform the heating function intended. The immersion heater shall be constructed of parts that are sound in structure and free from any repair.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale or mismatching. In no event shall such processes as welding, peening, plugging, or filling with solders, metallic or non-metallic pastes be used on castings or forgings to reclaim any defective parts for use on the heater.

3.3.2 Welding, brazing or soldering. Welding, brazing, or soldering shall be employed only where those operations are required in the original design. These operations shall not be employed as repair measures for defective parts.

3.3.3 Fastening devices. All screws, pins, bolts and similar fasteners shall be installed with means for preventing loss of tightness. Such fasteners subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.4 Surfaces. All surfaces of castings, forgings, molded parts, stampings and welded parts shall be cleaned and free from sand, dirt, fins, sprues, scale, flux or other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled unless sharpness is required to perform a necessary function.

3.4 Performance. The immersion heater shall be capable of performing the necessary functions of providing heat for the A/M32R-6 recharger and shall be capable of passing all tests in Section 4 of this specification.

3.5 Nameplate. A corrosion resistant metal nameplate marked in accordance with MIL-STD-130 shall be securely attached to each heater. The nameplate shall contain the information listed below.

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Nomenclature
Manufacturer's Name
Manufacturer's Model Designation
Manufacturer's Serial Number
Power Input Characteristics
Date of Manufacture
Contract or Order Number
Federal Stock Number
U.S.

3.6 Workmanship. Workmanship on the heater shall be commensurate with the requirements specified herein.

3.7 Fire and casualty hazards.

3.7.1 Proof of compliance. Each bidder shall submit to the contracting officer proof that the immersion heater he proposes to supply under this specification conforms to the applicable requirements of Underwriters' Laboratory Standards.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be performed on the first article items submitted in accordance with 3.1. First article inspection shall consist of the examination in 4.5 and tests in 4.6.2, 4.6.3, and 4.6.4. Failure of the first article to pass the examination or one or more first article tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Quality conformance inspection shall consist of the examination in 4.5 and 4.7, and the tests in 4.6.1.

4.5 Examination. Each immersion heater shall be examined for design, construction, materials, electrical equipment, and workmanship to determine compliance with the requirements of this specification.

4.6 Tests.

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4.6.1 Dielectric strength, dry. There shall be no flashover or breakdown of insulation when the heater is subjected to twice the rated voltage plus 500 volts. The test potential may be AC or DC of the same value and shall be applied for one minute between all terminals not in a common circuit and from all terminals to ground.

4.6.2 Water resistance test. The heater shall be submerged to a depth of three feet in tap water at a temperature of 75° ± 10° F and held there for 24 hours, and then removed. Examination after this test shall show that no water entered the heater or electrical connections and that operation of the heater has not been impaired.

4.6.3 Dielectric strength, wet. Immediately following the water resistance test, para 4.6.2, the heater shall be subjected to the dielectric tests as specified in para 4.6.1. The heater shall show no signs of flashover or breakdown of insulation in this test.

4.6.4 Reliability test. In an ambient temperature of approximately 40° F, the heater portion of the immersion heater shall be submerged in 50 gallons of water. The terminals and connections shall be enclosed in a compartment. An environment of approximately 120° F temperature and 90% relative humidity shall be maintained in the compartment for the duration of the test. The heater shall then be operated at rated capacity for a period of not less than 500 hours. The volume of water shall be maintained at approximately 50 gallons by addition to replace loss by evaporation. Any degradation which would impair the operation of the heater shall be cause for rejection.

4.7 Inspection of preparation for delivery. Preservation, packaging, packing, and marking for shipment and storage shall be inspected to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packing and packaging shall be Level A or C as specified (see 6.1).

5.1.1 Level A. Each heater shall be preserved and packaged in accordance with Method III of Specification MIL-P-116. Each terminal end shall be wrapped with commercial waterproof paper and sealed. A commercial fiberboard sleeve shall be placed over the terminal ends to prevent damage by distortion.

5.1.2 Level C. Preservation and packaging shall be in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be Level A, B, or C, as specified.

5.2.1 Level A. Each heater packaged as described in 5.1.1 shall be packed in an export type fiberboard container conforming to the requirements of Specifications PPP-B-591, PPP-B-636, or PPP-B-640. PPP-B-591, export type container, shall be provided with a waterproof case liner conforming to MIL-L-10547 and sealed in accordance with appendix thereto.

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5.2.2 Level B. Each heater packaged as described in 5.1.1, shall be packed in a domestic type fiberboard container conforming to the requirements of Specification PPP-B-591, PPP-B-636 or PPP-B-640.

5.2.3 Level C. Heaters shall be packed to insure carrier acceptance and safe delivery to destination in compliance with rules or regulations applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of Standard MIL-STD-129.

6. NOTES

6.1 Intended use. The immersion heater is intended for use in the converter of the A/M32R-6 Oxygen/Nitrogen Recharger.

6.2 Ordering data. Procurement data should specify the following:

a. Title, number, and date of this specification.

b. Level of preservation, packaging and packing required (see 5.1 and 5.2).

Custodian:
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Preparing activity:
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4540-F0036

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

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CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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